which will continue to expand infinitely. Surprisingly, current observations of the most distant galaxies suggest an accelerating expansion of the Universe instead.

2.6. QUARKS AND THE STANDARD MODEL

All particles are considered to be possible states in which matter can condense. These states are related to the force that forms them. In this sense the solar system is a state of gravitational force, an atom is a state of electromagnetic force, and a nucleon is a state of the strong interaction force. A particle can represent a positive energy state of a system while its analog antiparticle represents the negative state of the same system. Some regular patterns have been found for the elementary particles which indicate that many of them in fact may only be exited states of the same particle, differing in quantum numbers such as spin (or "hyper charge"); in fact hundreds of such states are now known. For example, the neutron has a mass corresponding to 939 MeV and spin 1/2, and there is a baryon with mass 1688 MeV and spin 5/2 with all its other properties like those of the neutron: the heavier particle must be a highly excited state of the neutron.

Though many attempts have been made to unify all particles into one simple theory, this did not succeed until the quark theory was developed. To explain this we have to go back somewhat in time.

The spin of a charged particle leads to the formation of a *magnetic moment* directed along the axis of "rotation". It was discovered in the late 1930s that the magnetic moment of the proton spin ($M_p = 1.41 \times 10^{-26}$ J T^{-1}) is about 1/700 of the electron spin ($M_e = 9.27 \times 10^{-24}$ J T^{-1}), although theory predicts a ratio of 1/1836 ($= M_p/M_e$; see Ch. 6). Also, the neutron has a negative magnetic moment ($M_n = -0.97 \times 10^{-26}$ J T^{-1}). The only explanation scientists could offer for this deviation was that the proton is not an evenly charged rotating sphere, but contains some "internal electrical currents", and also the neutron must contain some internal charges, which balance each other to appear uncharged. Thus, it was doubtful that protons and neutrons were truly elementary.

Around 1960 Hofstadter and co-workers at the large Stanford Linear Accelerator Center (SLAC, Ch. 16) proved that both the proton and the neutron have an uneven internal nuclear charge density. This came from studies of the scattering of high energy electrons ($\sim$1 GeV) against protons and neutrons. It was suggested by Gell-Mann that this could mean that the proton and neutron were composed of smaller particles, with fractional charge and mass, which he called *quarks*. The intense search for such particles (leading to the discovery of many new "elementary" particles) culminated in the late 1970s in experiments in which still higher electron energies (4 – 21 GeV) were used and the energy and scattering angle of the electrons measured. These revealed that the nucleons had hard internal scattering centers with charge 1/3 that of the electron and mass 1/3 that of the nucleon. These particles, quarks, are held together by *gluons*, which are carriers of the nuclear force.

These results have led to the *Standard Model* of the building blocks of matter. According to this model all matter on Earth — and likely in the Universe — (and including our own bodies) consists to > 99% of quarks with associated gluons. The rest is electrons.

Elementary particles now come in only two kinds: quarks and leptons. There are only six quarks and six leptons, see Table 2.3. The leptons are the electron, e, the muon, μ, and the *tauon* (tau particle), τ, and their respective neutrinos. The quarks and leptons are grouped together in *three families* (or generations) of two quarks and two leptons each. This makes 12 elementary building blocks, or 24 if one includes their anti particles; Table 2.3 only refers to our matter (i.e. koino matter). The leptons and quarks all have different properties and names, sometimes also referred to as *colors*. The physical theory relating these particles to each other is therefore named *Quantum Chromo Dynamics* (QCD).

All matter in nature belongs to the first family, which consist of two leptons, the electron and electron-neutrino, and the up-quark and the down-quark. The proton is made up of 2 up-quarks and 1 down-quark, giving it a charge of +1 and mass 1, while the neutron is made up of 1 up- and 2 down-quarks giving it a charge of 0 and mass of 1:

$$n = u^{+2/3} + d^{-1/3} + d^{-1/3} \tag{2.16}$$

$$p^{+} = u^{+2/3} + u^{+2/3} + d^{-1/3} \tag{2.17}$$

where u and d represent the up and down quarks, respectively. The neutron decay (§5.4.5) can be written according to the quark model: i.e. a d-quark is transformed into a u-quark with the simultaneous emission of an electron and an anti-neutrino, see Figure 2.3.

In all reactions the lepton number must be conserved: the total number of leptons minus anti leptons on each side of a decay or reaction process must be the same. A similar law is valid for the quarks. In the reaction above several quantum numbers are obeyed: (i) the charge is the same on both side, (ii) the lepton number is zero on both sides (none = electron minus anti neutrino), (iii) the quark number is conserved. The elementary reactions in Figure 2.2 can all be described in terms of lepton and quark transformations.

All hadrons contain 3 quarks, while all mesons are made up of 2 quarks or anti quarks. The quarks move around in the nucleus, which makes it difficult to observe these minute particles: if an atom had the size of the earth, the size of the quark would be about half a cm. The quarks cannot appear free but must appear together in groups of two or three.

The second family in Table 2.3 contains the "heavy electron", the muon and the muon neutrino, and the charm and the strange quarks. The third family contains the tau particle, the tau electron, and the two quarks referred to as top (or truth) and bottom (or beauty). These quarks can only be produced in high energy particle reactions.

By combination of quarks and leptons, the true elementary particles of nature, it is possible to systematize all known particles. The success of this theory, founded on good

Table 2.3 Classification and properties of elementary particles according to the standard model

Basic nature	Family	Name and symbol		Forces involved	Charge	Rest mass
Basic building blocks of nature	I	Leptons:	electron, e	EM,W	±1	0.511 MeV
			e-neutrino, ν_e	W	0	$<$18 eV
		Quarks:	up	EM,W,S	$+2/3$	1/3 u
			down	EM,W,S	$-1/3$	1/3 u
Formed in Big-Bang, cosmic rays and high-energy accelerators	II	Leptons:	muon, μ	EM,W	±1	105.6 MeV
			μ-neutrino, ν_μ	W	0	0?
		Quarks:	charm	EM,W,S		
			strange	EM,W,S		
	III	Leptons:	tau, τ	EM,W	±1	
			τ-neutrino,	W	0	0?
		Quarks:	top (or truth)	EM,W,S		
			bottom (or beauty)	EM,W,S		
Carriers of force:		Photon, γ		EM	0	0
bosons; spin = 0, 1, ..		Pion, π		S	0,±1	137 MeV
Pauli principle not valid		Gluon		EM,W,S		
		W^+,W^-,Z^0		W		

EM = electromagnetic force, W = weak interaction, S = strong interaction.

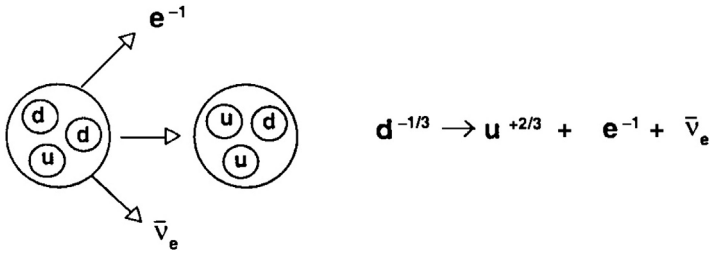


Figure 2.3 Neutron decay according to the quark model (simplified by omitting the intermediate vector boson, which decays to an electron and an antineutrino).

experimental evidence, has been so great that its name, the Standard Model of matter, is justified.

The force between two particles arises from the exchange of a "mediator" that carries the force at a finite speed: one of the particles emits the mediator, the other absorbs it. The mediator propagates through space and, briefly, is not lodged with either particle. These mediators have the same properties as the elementary particles — mass, electric charge, spin — so physicists often call them particles as well, even though their role in nature is quite different from that of the elementary particles. The mediators are the mortar that binds the particle building blocks together.

Three kinds of mediators — or exchange forces, as we have called them — are known in the nuclear world: photons which are involved in the electromagnetic force, gluons which are the mediators of the strong nuclear force, and the weak force mediators, which underlies the radioactive decay. We know now a great deal about the photons and the gluons, but little about the weak force. The weak force mediators are the W^+, W^- and Z^0 "particles". A high energy electron is supposed to be able to emit a Z^0, and then a positron can absorb the electron: the particles annihilate each other, leaving the Z^0 momentarily free. Afterwards the Z^0 must decay back into a pair of elementary particles, such as an electron and positron, or a quark and an antiquark. Z^0 particles are produced in high energy proton-antiproton colliders; around 1983 researchers at the CERN and SLAC laboratories were able to determine the Z^0 mass to 91.2 GeV. This is the heaviest known unit of matter today.

2.7. EXERCISES

2.1. Which type of mesons are released in high energy particle interactions, and why?

2.2. (a) What kinds of forces exist in nature? (b) How does the weak interaction manifest its properties?

2.3. (a) What are bosons and how do they differ from fermions? (b) Does the difference have any practical consequence?